

COMPUTER APPLICATIONS

CLASS X

There will be **one** written paper of **two hours** duration carrying 100 marks and Internal Assessment of 100 marks.

THEORY – 100 Marks

1. Revision of Class IX Syllabus

(i) Introduction to Object Oriented Programming concepts, (ii) Elementary Concept of Objects and Classes, (iii) Values and Data types, (iv) Operators in Java, (v) Input in Java, (vi) Mathematical Library Methods, (vii) Conditional constructs in Java, (viii) Iterative constructs in Java, (ix) Nested for loops.

2. Class as the Basis of all Computation

Objects and Classes

Objects encapsulate state and behaviour – numerous examples; member variables; attributes or features. Variables define state; member methods; Operations/methods/messages/ methods define behaviour.

Classes as abstractions for sets of objects; class as an object factory; primitive data types, composite data types. Variable declarations for both types; difference between the two types. Objects as instances of a class.

Consider real life examples for explaining the concept of class and object.

3. User - defined Methods

Need of methods, syntax of methods, forms of methods, method definition, method calling, method overloading, declaration of methods,

Ways to define a method, ways to invoke the methods – call by value [with programs] and call by reference [only definition with an example], Object creation - invoking the methods with respect to use of multiple methods with different names to implement modular programming, using data members and member methods, Actual parameters and formal parameters, Declaration of methods - static and non-static, method prototype / signature, - Pure and impure methods, - pass by value [with programs] and pass by reference [only definition with an example],

Returning values from the methods , use of multiple methods and more than one method with the same name (polymorphism - method overloading).

4. Constructors

Definition of Constructor, characteristics, types of constructors, use of constructors, constructor overloading.

Default constructor, parameterized constructor, constructor overloading., Difference between constructor and method.

5. Library classes

Introduction to wrapper classes, methods of wrapper class and their usage with respect to numeric and character data types. Autoboxing and Unboxing in wrapper classes.

Class as a composite type, distinction between primitive data type and composite data type or class types. Class may be considered as a new data type created by the user, that has its own functionality. The distinction between primitive and composite types should be discussed through examples. Show how classes allow user defined types in programs. All primitive types have corresponding class wrappers. Introduce Autoboxing and Unboxing with their definition and simple examples.

The following methods are to be covered:

*int parseInt(String s),
long parseLong(String s),
float parseFloat(String s),
double parseDouble(String s),
boolean isDigit(char ch),
boolean isLetter(char ch),
boolean isLetterOrDigit(char ch),
boolean isLowerCase(char ch),
boolean isUpperCase(char ch),
boolean isWhitespace(char ch),
char toLowerCase (char ch)
char toUpperCase(char ch)*

6. Encapsulation

Access specifiers and its scope and visibility.

Access specifiers – private, protected and public. Visibility rules for private, protected and public access specifiers. Scope of variables, class variables, instance variables, argument variables, local variables.

7. Arrays

Definition of an array, types of arrays, declaration, initialization and accepting data of single and double dimensional arrays, accessing the elements of single dimensional and double dimensional arrays.

Arrays and their uses, sorting techniques - selection sort and bubble sort; Search techniques – linear search and binary search, Array as a composite type, length statement to find the size of the array (sorting and searching techniques using single dimensional array only).

Declaration, initialization, accepting data in a double dimensional array, sum of the elements in row, column and diagonal elements [right and left], display the elements of two-dimensional array in a matrix format.

8. String handling

String class, methods of String class, implementation of String class methods, String array

The following String class methods are to be covered:

String trim ()

String toLowerCase()

String toUpperCase()

int length()

char charAt (int n)

int indexOf(char ch)

int lastIndexOf(char ch)

String concat(String str)

boolean equals (String str)

boolean equalsIgnoreCase(String str)

int compareTo(String str)

int compareToIgnoreCase(String str)

String replace (char oldChar,char newChar)

String substring (int beginIndex)

String substring (int beginIndex, int endIndex)

boolean startsWith(String str)

boolean endsWith(String str)

String valueOf(all types)

Programs based on the above methods, extracting and modifying characters of a string, alphabetical order of the strings in an array [Bubble and Selection sort techniques], searching for a string using linear search technique.

INTERNAL ASSESSMENT - 100 Marks

This segment of the syllabus is totally practical oriented. The accent is on acquiring basic programming skills quickly and efficiently.

Programming Assignments (Class X)

The students should complete a minimum of 20 laboratory assignments during the whole year to reinforce the concepts studied in class.

Suggested list of Assignments:

The laboratory assignments will form the bulk of the course. Good assignments should have problems which require design, implementation and testing. They should also embody one or more concepts that have been discussed in the theory class. A significant proportion of the time has to be spent in the laboratory. Computing can only be learnt by doing.

The teacher-in-charge should maintain a record of all the assignments done by the student throughout the year and give it due credit at the time of cumulative evaluation at the end of the year.

Some sample problems are given below as examples. The problems are of varying levels of difficulty:

- (i) User defined methods
 - (a) Programs depicting the concept of pure, impure, static, non- static methods.
 - (b) Programs based on overloaded methods.
 - (c) Programs involving data members, member methods invoking the methods with respect to the object created.

- (ii) Constructors
 - (a) Programs based on different types of constructors mentioned in the scope of the syllabus.
 - (b) Programs / outputs based on constructor overloading
- (iii) Library classes
 - (a) Outputs based on all the methods mentioned in the scope of the syllabus.
 - (b) Programs to check whether a given character is an uppercase/ lowercase / digit etc.
- (iv) Encapsulation

Questions based on identifying the different variables like local, instance, arguments, private, public, class variable etc.
- (v) Arrays
 - (a) Programs based on accessing the elements of an array.
 - (b) Programs based on sort techniques mentioned in the scope of the syllabus.
 - (c) Programs based on search techniques mentioned in the scope of the syllabus.
 - (d) Programs on Double dimensional arrays as given in the scope of the syllabus.
- (vi) String handling
 - (a) Outputs based on all the string methods mentioned in the scope of the syllabus.
 - (b) Programs based on extracting the characters from a given string and manipulating the same.
 - (c) Palindrome string, pig Latin, alphabetical order of characters, etc.

Important: This list is indicative only. Teachers and students should use their imagination to create innovative and original assignments.

EVALUATION

The teacher-in-charge shall evaluate all the assignments done by the student throughout the year [both written and practical work]. He/she shall ensure that most of the components of the syllabus have been used appropriately in the assignments. Assignments should be with appropriate list of variables and comment statements. The student has to mention the output of the programs.

Proposed Guidelines for Marking

The teacher should use the criteria below to judge the internal work done. Basically, four criteria are being suggested: class design, coding and documentation, variable description and execution or output. The actual grading will be done by the teacher based on his/her judgment. However, one possible way: divide the outcome for each criterion into one of 4 groups: excellent, good, fair/acceptable, poor/unacceptable, then use numeric values for each grade and add to get the total.

Class design:

Has a suitable class (or classes) been used?
 Are all attributes with the right kinds of types present?
 Is encapsulation properly done?
 Is the interface properly designed

Coding and documentation:

Is the coding done properly? (Choice of names, no unconditional jumps, proper organization of conditions, proper choice of loops, error handling, code layout) Is the documentation complete and readable? (class documentation, variable documentation, method documentation, constraints, known bugs - if any).

Variable description:

Format for variable description:

Name of the Variable	Data Type	Purpose/description

Execution or Output:

Does the program run on all sample input correctly?

Evaluation of practical work will be done as follows:

Subject Teacher (Internal Examiner)		50 marks		
External Examiner		50 marks		
Criteria (Total- 50 marks)	Class design (10 marks)	Variable description (10 marks)	Coding and Documentation (10 marks)	Execution OR Output (20 marks)
Excellent	10	10	10	20
Good	8	8	8	16
Fair	6	6	6	12
Poor	4	4	4	8

An External Examiner shall be nominated by the Head of the School and may be a teacher from the faculty, but not teaching the subject in the relevant section/class. For example, A teacher of Computer Science of class VIII may be deputed to be the External Examiner for class X.

The total marks obtained out of 100 are to be sent to CISCE by the Head of the school.

The Head of the school will be responsible for the online entry of marks on CISCE's CAREERS portal by the due date.

EQUIPMENT

There should be enough computer systems to provide for a teaching schedule where at least three-fourth of a time available is used for programming and assignments/practical work. The course shall require at least 4 periods of about 40 minutes duration per week. In one week, out of 4 periods the time should be divided as follows:

- 2 periods – Lecture cum demonstration by the Instructor.
- 2 periods – Assignments/Practical work.

The hardware and software platforms should be such that students can comfortably develop and run programs on those machines.

Since hardware and software evolve and change very rapidly the schools shall need to upgrade them as required. Following are the minimal specifications as of now.

RECOMMENDED FACILITIES:

- A lecture cum demonstration room with a MULTIMEDIA PROJECTOR/ an LCD and Overhead Projector (OHP) attached to the computer.
- A white board with white board markers should be available.
- A fully equipped Computer Laboratory that allows one computer per student.
- The computers should have a minimum of 1 GB RAM and at least a P - IV or Equivalent Processor.
- Good Quality printers.
- A scanner, a web cam/a digital camera (Should be provided if possible).

SOFTWARE FOR CLASSES IX & X

Any suitable Operating System can be used.

For teaching fundamental concepts of computing using object oriented approach, Blue J environment (3.2 or higher version) compatible with JDK (5.0 or higher version) as the base or any other editor or IDE, compatible with JDK (5.0 or higher version) as the base may be used. Ensure that the latest versions of software are used.